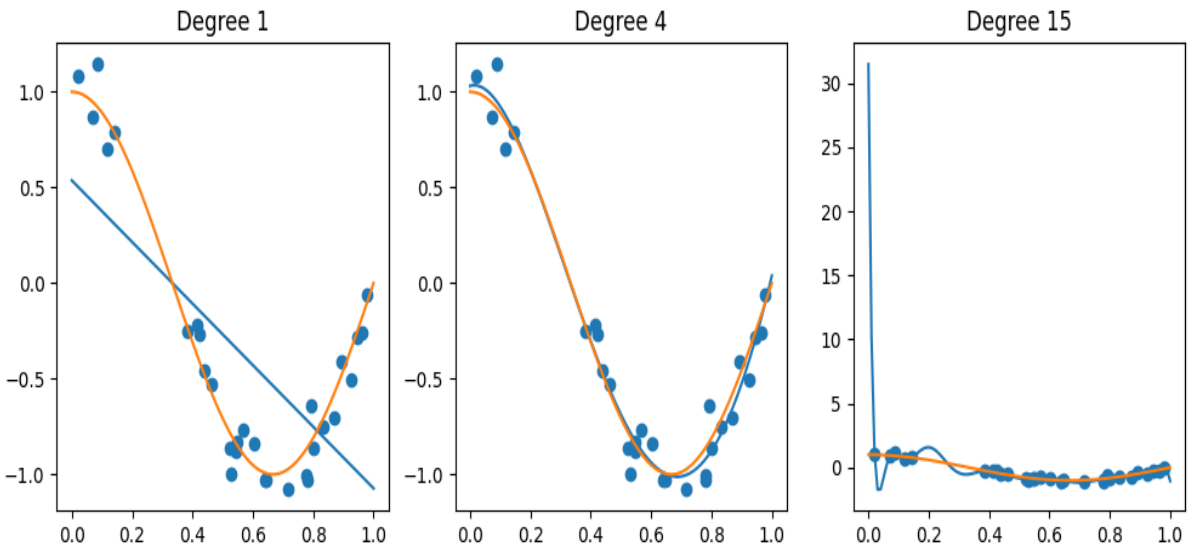
Experiment 4:

Aim: To analyse the performance of overfitting using Python.

Code:

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.preprocessing import PolynomialFeatures
from sklearn.linear_model import LinearRegression
from sklearn.pipeline import make_pipeline
def true_fun(x):
    return np.cos(1.5 * np.pi * x)
np.random.seed(0)
X = np.sort(np.random.rand(30))
y = true_fun(X) + np.random.randn(30) * 0.1
degrees = [1, 4, 15]
plt.figure(figsize=(12, 4))
for i in range(3):
    model = make_pipeline(
        PolynomialFeatures(degrees[i]),
        LinearRegression()
    )
    model.fit(X.reshape(-1, 1), y)
    X_test = np.linspace(0, 1, 100)
    y_pred = model.predict(X_test.reshape(-1, 1))
    plt.subplot(1, 3, i + 1)
    plt.scatter(X, y)
    plt.plot(X_test, y_pred)
    plt.plot(X_test, true_fun(X_test))
    plt.title("Degree " + str(degrees[i]))
plt.show() j
```

Output:



Result: Thus, the performance of overfitting was successfully demonstrated using polynomial regression.